

Depen Morwani

+1 (857)928-6485 • depenmorwani@gmail.com • depenm.github.io •

RESEARCH INTERESTS

Deep Learning Optimization, Science of Deep Learning

EDUCATION

PhD, <i>Computer Science</i>	Aug 2022 - May 2026 (Expected)
Harvard University, USA	GPA: 4.0/4.0
Masters with Thesis, <i>Computer Science</i>	Jul 2019 - May 2021
IIT Madras, India	GPA: 9.82/10
Bachelors with Honors, <i>Computer Science</i>	Jul 2012 - May 2016
IIT Bombay, India	GPA: 8.74/10

PUBLICATIONS

- **The Potential of Second-Order Optimization for LLMs: A Study with Full Gauss-Newton**, *International Conference on Learning Representations (ICLR)*, 2026
Natalie Abreu, Nikhil Vyas, Sham Kakade, [Depen Morwani](#) [arXiv](#)
- **Seesaw: Accelerating Training by Balancing Learning Rate and Batch Size Scheduling**, *International Conference on Learning Representations (ICLR)*, 2026
Alexandru Meterez, [Depen Morwani](#), Jingfeng Wu, Costin-Andrei Oncescu, Cengiz Pehlevan, Sham Kakade [arXiv](#)
- **SOAP: Improving and Stabilizing Shampoo using Adam**, *International Conference on Learning Representations (ICLR)*, 2025
Nikhil Vyas, [Depen Morwani](#), Rosie Zhao, Itai Shapira, David Brandfonbrener, Lucas Janson, Sham Kakade [arXiv](#)
- **A New Perspective on Shampoo's Preconditioner**, *International Conference on Learning Representations (ICLR)*, 2025
[Depen Morwani](#), Itai Shapira, Nikhil Vyas, Eran Malach, Sham Kakade, Lucas Janson [arXiv](#)
- **Deconstructing What Makes a Good Optimizer for Language Models**, *International Conference on Learning Representations (ICLR)*, 2025
Rosie Zhao, [Depen Morwani](#), David Brandfonbrener, Nikhil Vyas, Sham Kakade [arXiv](#)
- **How Does Critical Batch Size Scale in Pre-training?**, *International Conference on Learning Representations (ICLR)*, 2025
Hanlin Zhang, [Depen Morwani](#), Nikhil Vyas, Jingfeng Wu, Difan Zou, Udaya Ghai, Dean Foster, Sham Kakade [arXiv](#)
- **Feature emergence via margin maximization: case studies in algebraic tasks**, *Spotlight (5% acceptance rate) at International Conference on Learning Representations (ICLR)*, 2024
[Depen Morwani](#), Benjamin L. Edelman, Costin-Andrei Oncescu, Rosie Zhao, Sham Kakade [arXiv](#)
- **Beyond Implicit Bias: The Insignificance of SGD Noise in Online Learning**, *Spotlight (3.5% acceptance rate) at International Conference on Machine Learning (ICML)*, 2024
Nikhil Vyas, [Depen Morwani](#), Rosie Zhao, Gal Kaplun, Sham Kakade, Boaz Barak [arXiv](#)
- **Simplicity Bias in 1-Hidden Layer Neural Networks**, *Neural Information Processing Systems (NeurIPS)*, 2023
[Depen Morwani](#), Jatin Batra, Prateek Jain, Praneeth Netrapalli [arXiv](#)
- **Feature-Learning Networks Are Consistent Across Widths At Realistic Scales**, *Neural Information Processing Systems (NeurIPS)*, 2023
Nikhil Vyas, Alex Atanasov, Blake Bordelon, [Depen Morwani](#), Sabarish Sainathan, Cengiz Pehlevan [arXiv](#)
- **Inductive Bias of Gradient Descent for Weight Normalized Smooth Homogeneous Neural Nets**, *Algorithmic Learning Theory (ALT)*, 2022
[Depen Morwani](#), Harish G. Ramaswamy [arXiv](#)

PREPRINTS

- **LOTION: Smoothing the Optimization Landscape for Quantized Training**, *Under submission at International Conference on Machine Learning (ICML)*, 2026

Mujin Kwun, [Depen Morwani](#), Chloe Huangyuan Su, Stephanie Gil, Nikhil Anand, Sham Kakade [arXiv](#)

- **Adam or Gauss-Newton? A Comparative Study In Terms of Basis Alignment and SGD Noise**, *Under submission at International Conference on Machine Learning (ICML), 2026*

Bingbin Liu, Rachit Bansal, [Depen Morwani](#), Nikhil Vyas, David Alvarez-Melis, Sham M. Kakade [arXiv](#)

- **Connections between Schedule-Free Optimizers, AdEMAMix, and Accelerated SGD Variants**, *Under submission at International Conference on Machine Learning (ICML), 2026*

[Depen Morwani](#), Nikhil Vyas, Hanlin Zhang, Sham Kakade [arXiv](#)

- **A Simplified Analysis of SGD for Linear Regression with Weight Averaging**, *Workshop Optimization for Machine Learning (OPT), 2025*

Alexandru Meterez, [Depen Morwani](#), Costin-Andrei Oncescu, Jingfeng Wu, Cengiz Pehlevan, Sham Kakade [arXiv](#)

- **AdaMeM: Memory Efficient Momentum for Adafactor**, *Workshop on Advancing Neural Network Training (WANT), 2024*
Nikhil Vyas, [Depen Morwani](#), Sham M. Kakade [openreview](#)

RESEARCH INTERNSHIP

Google DeepMind, London | *Research Intern* Apr 2025 - Sep 2025

- Developing a theoretical understanding of the interaction between layer alignment and progressive sharpening, and proposing a regularizer for mitigating sharpening.

Microsoft Research, NYC | *Research Intern* Jun 2024 - Aug 2024

- Understanding the interaction between learning rate decay, weight averaging and edge of stability.

Google AI, India | *Research Associate - Deep Learning* Jun 2021 - Jul 2022

- Rigorously defined and established the notion of **simplicity bias** for 1-hidden layer neural networks. [arXiv](#)
- Proposed and verified the efficacy of *orthogonal projection* training method for mitigating simplicity bias on real-world datasets.

IIT Madras, India | *Project Associate - Reinforcement Learning* Aug 2018 - Jun 2019

- Modeled the behavior of Taxi Driver as a sequential decision making problem, and quantified the value of demand information received by the driver by cleverly combining **Monte-Carlo sampling** and **Value Iteration**. [report](#)

INDUSTRY EXPERIENCE

Goldman Sachs, India | *Financial Analyst - Short Term Rates Desk* Jun 2016 - Jul 2018

- Responsible for **pricing and risk** calculation of various rates products such as Swaps and Futures.
- Improved yield curve fitting by modeling it as a **Bayesian Inference** problem with Gaussian process prior and optimized the implementation by expressing the posterior as a sum of basis functions.

Samsung Electronics, S. Korea | *Summer Intern - Machine Learning* May 2015 - Jul 2015

- Predicted upcoming failure in Base Station, which is an important component of mobile networks.
- Increased the precision of the **Random Forest** Algorithm from 70% to 90% at a recall of 60%, by designing better features.

TEACHING EXPERIENCE

- Fall 2023 Teaching Fellow for **Algorithms for Data Science** (CS224) at Harvard University
- Spring 2019 and Spring 2020 Co-Instructor and Teaching Assistant for **Big Data Laboratory**(CS4830) at IIT Madras
- Fall 2019 Teaching Assistant for **Non-Linear Optimization**(CS5020) at IIT Madras

ACHIEVEMENTS

- Organized the **High-dimensional Learning Dynamics** (HiLD) workshop at ICML 2024.
- Secured **5th** position in the **Predicting Generalization in Deep Learning** (PGDL) competition held at *Neural Information Processing Systems (NeurIPS) 2020* [arXiv](#)

INTERESTS AND SKILLS

- *Languages* - C++, Python
- *Frameworks* - Pytorch, JAX
- *Interests* - Optimization, Probability, Statistical Learning Theory